Noora Health Data Engineer Assignment

Data Extraction and Loading

Database Setup and Data Import

```
import pandas as pd
from sqlalchemy import create_engine

# Database connection setup
engine = create_engine("mysql+pymysql://root:pass123@localhost/nura")

# Load Excel data from the provided file
excel_file = r"C:\Users\ASUS\Downloads\Data Engineer Task Assignment.xlsx"
sheets = pd.read_excel(excel_file, sheet_name=None)

# Import each sheet into database tables
for sheet_name, df in sheets.items():
    table_name = sheet_name.lower().replace(" ", "_")
    df.to_sql(table_name, engine, if_exists="replace", index=False)
    print(f"Imported sheet '{sheet_name}' into table '{table_name}'")
```

Data Verification

```
-- Verify data import
SELECT * FROM nura.messages LIMIT 5;
SELECT * FROM nura.statuses LIMIT 5;
```

Data Transformation

Creating Unified Messages Table

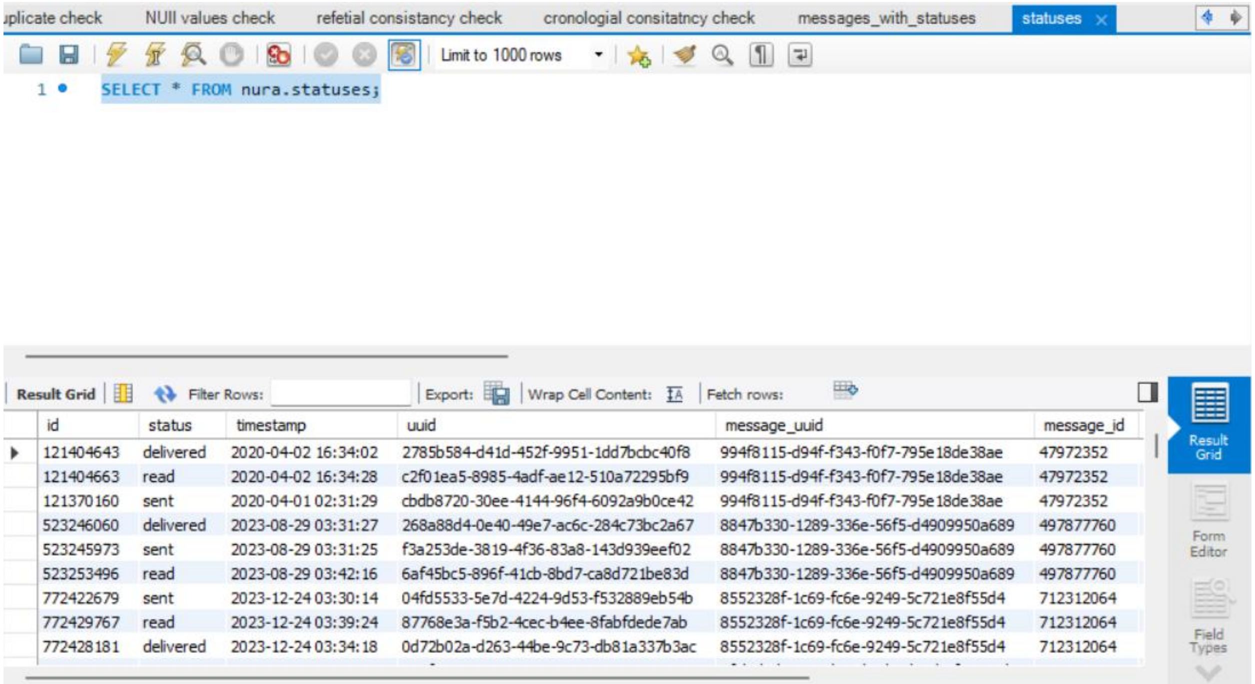
```
CREATE TABLE nura.messages_with_statuses AS
SELECT
    m.id AS message_id,
    m.message_type,
    m.masked_addressees,
    m.masked_author,
    m.content,
    m.author_type,
    m.direction,
    m.external_id,
    m.external_timestamp,
    m.masked_from_addr,
    m.is_deleted,
    m.last_status,
    m.last_status_timestamp,
    m.rendered_content,
    m.source_type,
    m.uuid AS message_uuid,
    m.inserted_at AS message_inserted_at,
    m.updated_at AS message_updated_at,
    s.id AS status_id,
```

```sql
    s.status,
    s.timestamp AS status_timestamp,
    s.uuid AS status_uuid,
    s.message_uuid AS status_message_uuid,
    s.message_id AS status_message_id,
    s.number_id,
    s.inserted_at AS status_inserted_at,
    s.updated_at AS status_updated_at
FROM nura.messages m
LEFT JOIN nura.statuses s ON m.id = s.message_id;
```

importdata.py ● nura_vizulization.py 1

C: > Users > ASUS > Documents > Python Scripts > importdata.py > ...

```python
import pandas as pd
from sqlalchemy import create_engine


engine = create_engine("mysql+pymysql://root:pass123@localhost/nura")

excel_file = r"C:\Users\ASUS\Downloads\Data Engineer Task Assignment.xlsx"


sheets = pd.read_excel(excel_file, sheet_name=None)


for sheet_name, df in sheets.items():
    table_name = sheet_name.lower().replace(" ", "_")  # clean table name
    df.to_sql(table_name, engine, if_exists="replace", index=False)
    print(f"Imported sheet '{sheet_name}' into table '{table_name}'")
```



uplicate check | NUll values check | refetial consistancy check | cronologial consitatncy check | messages_with_statuses | statuses

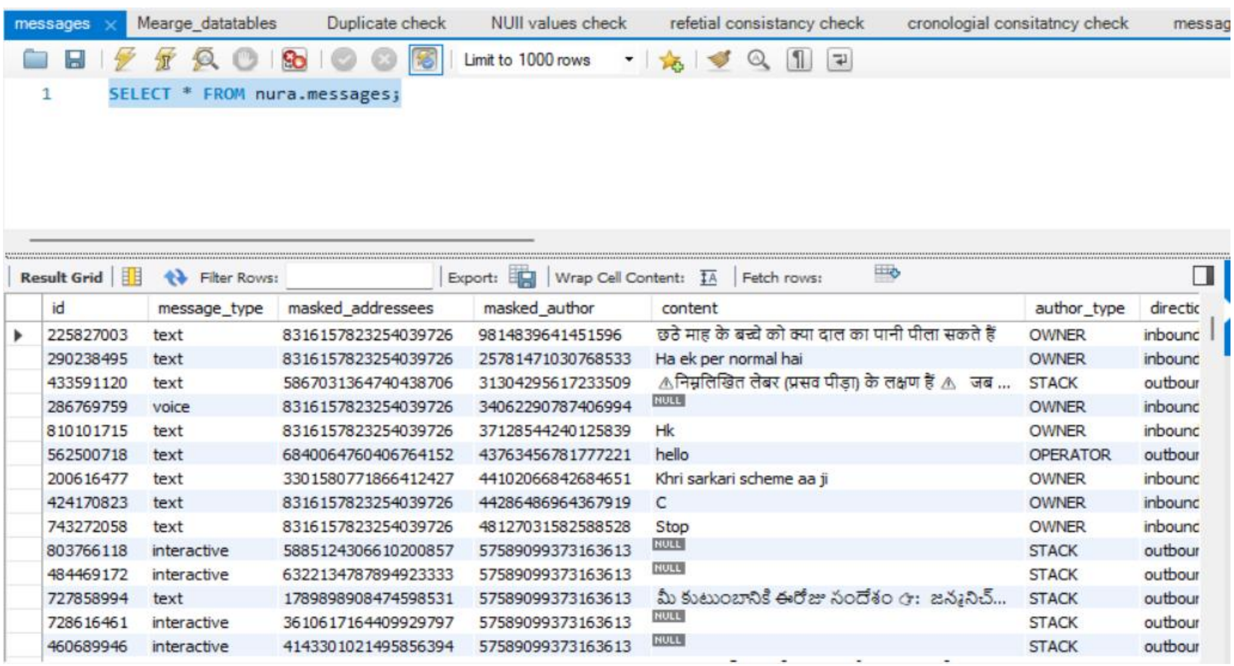
Limit to 1000 rows

```sql
1 SELECT * FROM nura.statuses;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

id	status	timestamp	uuid	message_uuid	message_id
121404643	delivered	2020-04-02 16:34:02	2785b584-d41d-452f-9951-1dd7bcbc40f8	994f8115-d94f-f343-f0f7-795e18de38ae	47972352
121404663	read	2020-04-02 16:34:28	c2f01ea5-8985-4adf-ae12-510a72295bf9	994f8115-d94f-f343-f0f7-795e18de38ae	47972352
121370160	sent	2020-04-01 02:31:29	cbdb8720-30ee-4144-96f4-6092a9b0ce42	994f8115-d94f-f343-f0f7-795e18de38ae	47972352
523246060	delivered	2023-08-29 03:31:27	268a88d4-0e40-49e7-ac6c-284c73bc2a67	8847b330-1289-336e-56f5-d4909950a689	497877760
523245973	sent	2023-08-29 03:31:25	f3a253de-3819-4f36-83a8-143d939eef02	8847b330-1289-336e-56f5-d4909950a689	497877760
523253496	read	2023-08-29 03:42:16	6af45bc5-896f-41cb-8bd7-ca8d721be83d	8847b330-1289-336e-56f5-d4909950a689	497877760
772422679	sent	2023-12-24 03:30:14	04fd5533-5e7d-4224-9d53-f532889eb54b	8552328f-1c69-fc6e-9249-5c721e8f55d4	712312064
772429767	read	2023-12-24 03:39:24	87768e3a-f5b2-4cec-b4ee-8fabfdede7ab	8552328f-1c69-fc6e-9249-5c721e8f55d4	712312064
772428181	delivered	2023-12-24 03:34:18	0d72b02a-d263-44be-9c73-db81a337b3ac	8552328f-1c69-fc6e-9249-5c721e8f55d4	712312064

Result Grid | Form Editor | Field Types



messages | Mearge_datatables | Duplicate check | NUll values check | refetial consistancy check | cronologial consitatncy check | messag

Limit to 1000 rows

```sql
1 SELECT * FROM nura.messages;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

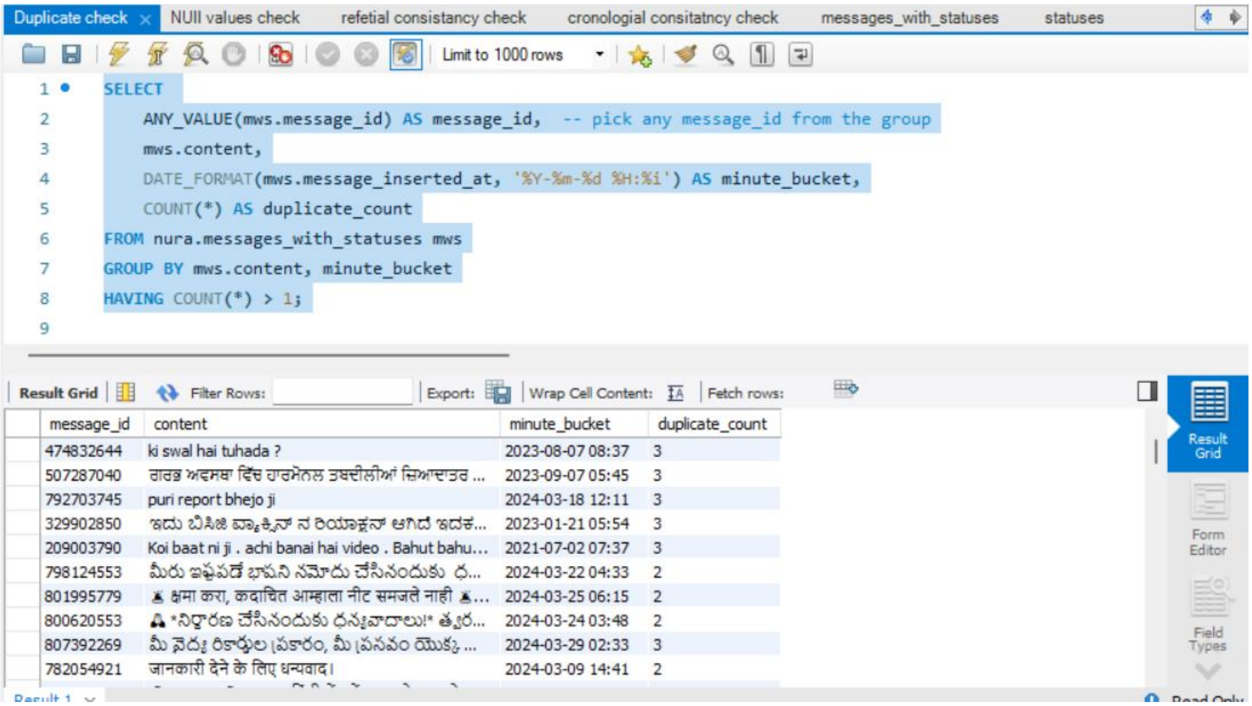
id	message_type	masked_addressees	masked_author	content	author_type	directic
225827003	text	8316157823254039726	9814839641451596	छठे माह के बच्चे को क्या दाल का पानी पीला सकते हैं	OWNER	inbounc
290238495	text	8316157823254039726	25781471030768533	Ha ek per normal hai	OWNER	inbounc
433591120	text	5867031364740438706	31304295617233509	⚠निम्नलिखित लेबर (प्रसव पीड़ा) के लक्षण हैं ⚠ जब ...	STACK	outbour
286769759	voice	8316157823254039726	34062290787406994	NULL	OWNER	inbounc
810101715	text	8316157823254039726	37128544240125839	Hk	OWNER	inbounc
562500718	text	6840064760406764152	43763456781777221	hello	OPERATOR	outbour
200616477	text	3301580771866412427	44102066842684651	Khri sarkari scheme aa ji	OWNER	inbounc
424170823	text	8316157823254039726	44286486964367919	C	OWNER	inbounc
743272058	text	8316157823254039726	48127031582588528	Stop	OWNER	inbounc
803766118	interactive	5885124306610200857	57589099373163613	NULL	STACK	outbour
484469172	interactive	6322134787894923333	57589099373163613	NULL	STACK	outbour
727858994	text	1789898908474598531	57589099373163613	మీ కుటుంబానికి ఈరోజు సందేశం ః: జన్మనిచ్...	STACK	outbour
728616461	interactive	3610617164409929797	57589099373163613	NULL	STACK	outbour
460689946	interactive	4143301021495856394	57589099373163613	NULL	STACK	outbour

Data Quality Checks

1. Duplicate Detection

```sql
SELECT
   ANY_VALUE(mws.message_id) AS message_id,
   mws.content,
   DATE_FORMAT(mws.message_inserted_at, '%Y-%m-%d %H:%i') AS minute_bucket,
   COUNT(*) AS duplicate_count
FROM nura.messages_with_statuses mws
```

GROUP BY mws.content, minute_bucket
HAVING COUNT(*) > 1;

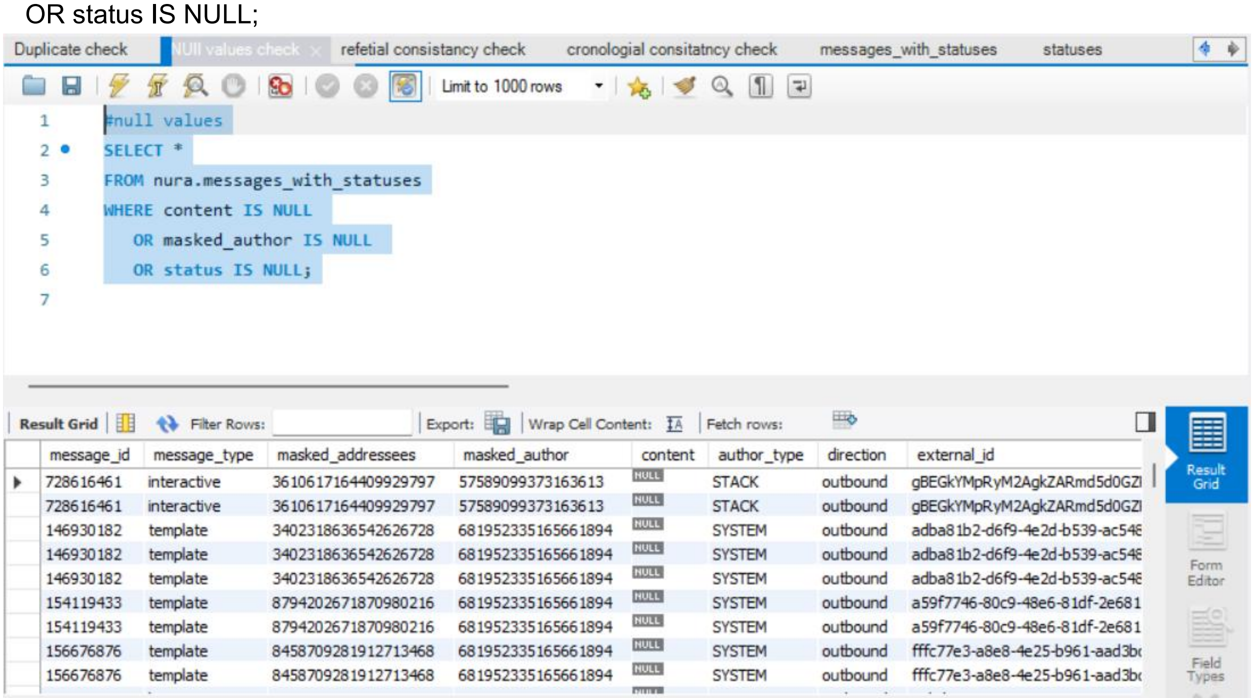


```
SELECT
    ANY_VALUE(mws.message_id) AS message_id,  -- pick any message_id from the group
    mws.content,
    DATE_FORMAT(mws.message_inserted_at, '%Y-%m-%d %H:%i') AS minute_bucket,
    COUNT(*) AS duplicate_count
FROM nura.messages_with_statuses mws
GROUP BY mws.content, minute_bucket
HAVING COUNT(*) > 1;
```

message_id	content	minute_bucket	duplicate_count
474832644	ki swal hai tuhada ?	2023-08-07 08:37	3
507287040	ਗਰਭ ਅਵਸਥਾ ਵਿੱਚ ਹਾਰਮੋਨਲ ਤਬਦੀਲੀਆਂ ਜ਼ਿਆਦਾਤਰ ...	2023-09-07 05:45	3
792703745	puri report bhejo ji	2024-03-18 12:11	3
329902850	ಇದು ಬಿಸಿಜಿ ವ್ಯಾಕ್ಸಿನ್ ನ ರಿಯಾಕ್ಷನ್ ಆಗಿದೆ ಇದಕ...	2023-01-21 05:54	3
209003790	Koi baat ni ji . achi banai hai video . Bahut bahu...	2021-07-02 07:37	3
798124553	మీరు ఇష్టపడే భాషని నమోదు చేసినందుకు ధ...	2024-03-22 04:33	2
801995779	क्षमा करा, कदाचित आम्हाला नीट समजले नाही ...	2024-03-25 06:15	2
800620553	*నిర్ధారణ చేసినందుకు ధన్యవాదాలు!* త్వర...	2024-03-24 03:48	2
807392269	మీ వైద్య రికార్డుల ప్రకారం, మీ ప్రసవం యొక్క ...	2024-03-29 02:33	3
782054921	जानकारी देने के लिए धन्यवाद।	2024-03-09 14:41	2

2. Null Value Check

SELECT *
FROM nura.messages_with_statuses
WHERE content IS NULL
   OR masked_author IS NULL
   OR status IS NULL;

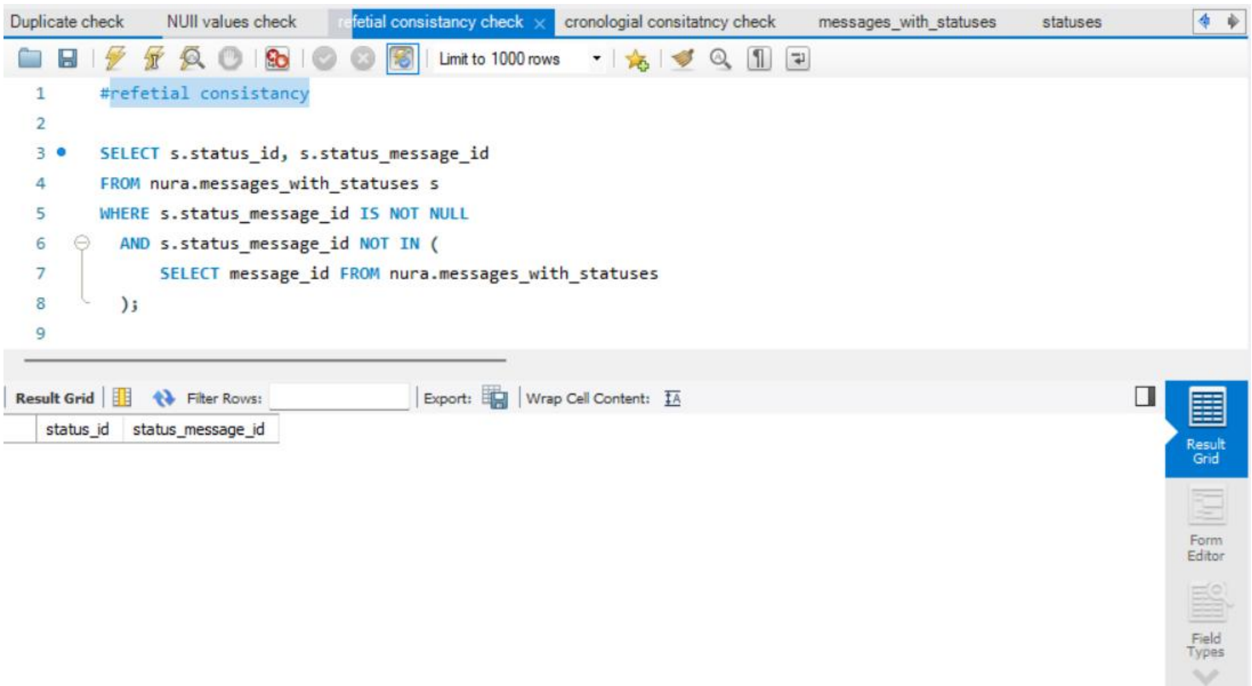


```
#null values
SELECT *
FROM nura.messages_with_statuses
WHERE content IS NULL
   OR masked_author IS NULL
   OR status IS NULL;
```

message_id	message_type	masked_addressees	masked_author	content	author_type	direction	external_id
728616461	interactive	361061716440992979 7	5758909937316361 3	NULL	STACK	outbound	gBEGkYMpRyM2AgkZARmd5d0GZI
728616461	interactive	3610617164409929797	57589099373163613	NULL	STACK	outbound	gBEGkYMpRyM2AgkZARmd5d0GZI
146930182	template	340231863654262672 8	681952335165661894	NULL	SYSTEM	outbound	adba81b2-d6f9-4e2d-b539-ac548
146930182	template	3402318636542626728	681952335165661894	NULL	SYSTEM	outbound	adba81b2-d6f9-4e2d-b539-ac548
146930182	template	3402318636542626728	681952335165661894	NULL	SYSTEM	outbound	adba81b2-d6f9-4e2d-b539-ac548
154119433	template	8794202671870980216	681952335165661894	NULL	SYSTEM	outbound	a59f7746-80c9-48e6-81df-2e681
154119433	template	8794202671870980216	681952335165661894	NULL	SYSTEM	outbound	a59f7746-80c9-48e6-81df-2e681
156676876	template	8458709281912713468	681952335165661894	NULL	SYSTEM	outbound	fffc77e3-a8e8-4e25-b961-aad3b
156676876	template	8458709281912713468	681952335165661894	NULL	SYSTEM	outbound	fffc77e3-a8e8-4e25-b961-aad3b

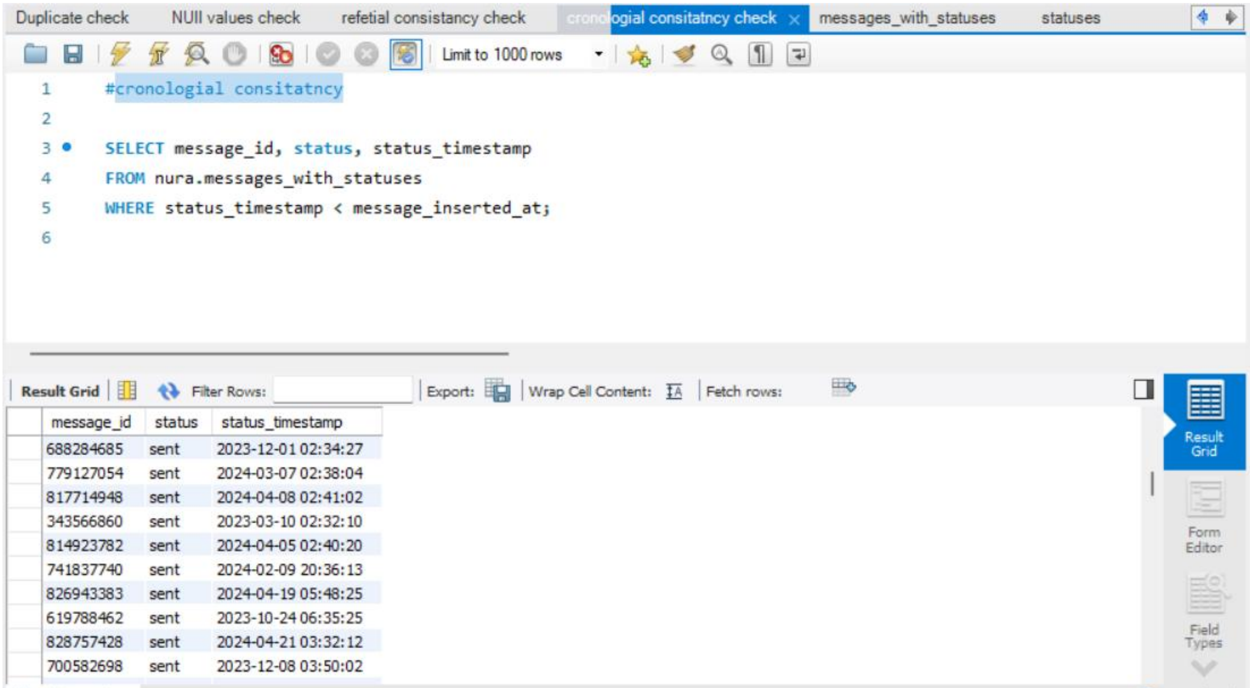
3. Referential Integrity Check

```
SELECT s.status_id, s.status_message_id
FROM nura.messages_with_statuses s
WHERE s.status_message_id IS NOT NULL
  AND s.status_message_id NOT IN (
      SELECT message_id FROM nura.messages_with_statuses
  );
```



4. Chronological Consistency Check

```
SELECT message_id, status, status_timestamp
FROM nura.messages_with_statuses
WHERE status_timestamp < message_inserted_at;
```



Data Visualization

Python Setup and Database Connection

import pandas as pd
import plotly.express as px
import pymysql
from sqlalchemy import create_engine
from datetime import datetime, timedelta

# Database connection
user = 'root'
password = 'pass123'
host = 'localhost'
port = 3306
database = 'nura'

engine = create_engine(f'mysql+pymysql://{user}:{password}@{host}:{port}/{database}')

# Load transformed data
df = pd.read_sql('SELECT * FROM messages_with_statuses', con=engine)

Data preprocessing
df['sent_at'] = pd.to_datetime(df['message_inserted_at'])
df['read_at'] = pd.to_datetime(df['status_timestamp'])
df['week'] = df['sent_at'].dt.to_period('W').apply(lambda r: r.start_time)
Visualization 1: Total and Active Users Over Time
total_users = df.groupby('week')['author_type'].nunique().reset_index(name='total_users')

```
active_users = df[df['direction'] ==
'inbound'].groupby('week')['author_type'].nunique().reset_index(name='active_users')

users_over_time = pd.merge(total_users, active_users, on='week')

fig1 = px.line(users_over_time, x='week', y=['total_users', 'active_users'],
              title='Total and Active Users Over Time')
fig1.show()
```

Visualization 2: Read Rate Analysis

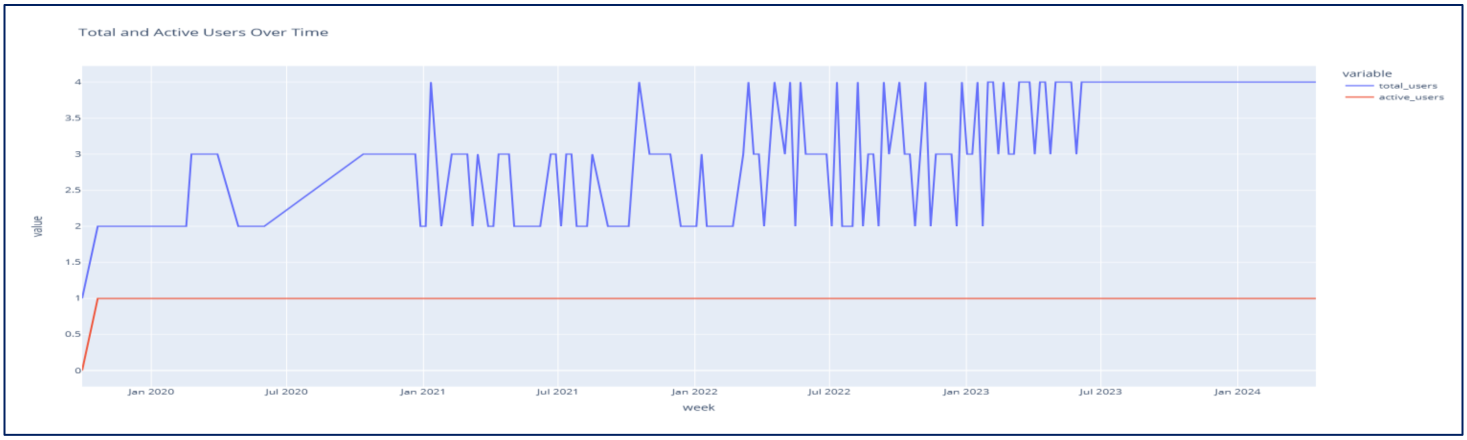
```
outbound = df[(df['direction'] == 'outbound') & (df['status'] != 'failed')]
read_fraction = outbound['read_at'].notna().mean()

print(f"Fraction of outbound messages read: {read_fraction:.2%}")

# Weekly read rates
read_by_week = outbound.groupby('week')['read_at'].apply(lambda x: x.notna().mean()).reset_index()

fig2 = px.bar(read_by_week, x='week', y='read_at',
             labels={'read_at': 'Fraction Read'},
             title='Fraction of Outbound Messages Read Over Time')
fig2.show()
```

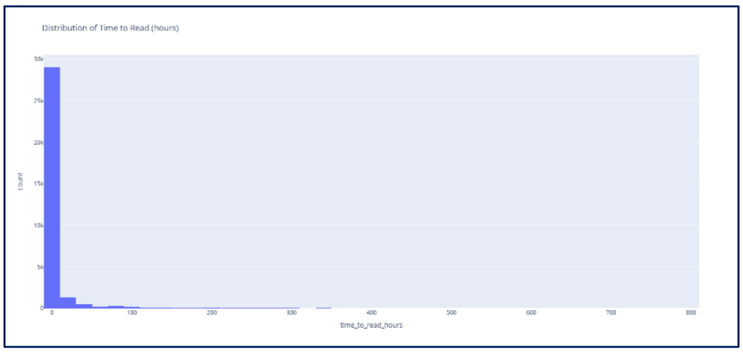
Visualization 3: Time to Read Distribution

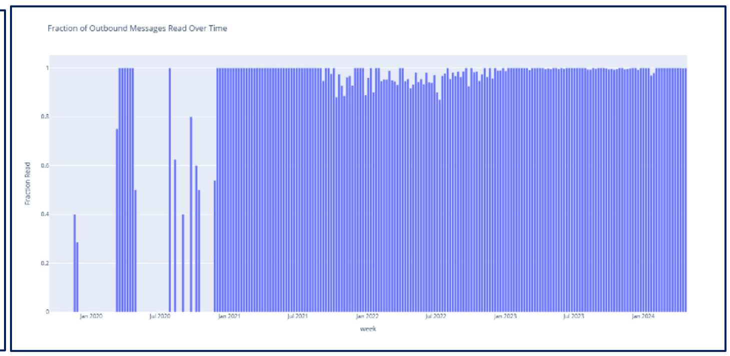
```
read_outbound = outbound[outbound['read_at'].notna()].copy()
read_outbound['time_to_read_hours'] = (read_outbound['read_at'] -
read_outbound['sent_at']).dt.total_seconds() / 3600

fig3 = px.histogram(read_outbound, x='time_to_read_hours', nbins=50,
                   title='Distribution of Time to Read (hours)')
fig3.show()
```

Visualization 4: Recent Message Status Distribution

```
last_week_start = datetime.now() - timedelta(days=7)
last_week_msgs = df[(df['direction'] == 'outbound') & (df['sent_at'] >= last_week_start)]

status_counts = last_week_msgs['status'].value_counts().reset_index()
status_counts.columns = ['status', 'count']

fig4 = px.bar(status_counts, x='status', y='count',
             title='Outbound Messages in Last Week by Status')
fig4.show()
```







Summary

This analysis provides a comprehensive view of the WhatsApp messaging data including:

1. **Data Integration**: Successfully merged messages and statuses tables while preserving all information
2. **Quality Assurance**: Implemented multiple validation checks for data integrity
3. **User Engagement**: Tracked both total and active user metrics over time
4. **Message Performance**: Analyzed read rates and response times for outbound messages
5. **Recent Activity**: Monitored current message status distribution

The pipeline demonstrates effective ETL processes, data validation techniques, and meaningful visualization of messaging platform analytics.